Lab 7: Fuzzing for Input Validation Bugs

Module under test: self_storage.py (parse_rental, duration_average, payable_rent)
Tools: Python 3.13, pytest 9.1, Hypothesis 6.170
Repo: L6_L7_51_self_storage / Lab 7

Note: the outputs below are the exact terminal output of each command, rendered into this PDF.

Bugs found (fuzzing + property tests) and fixed:

1. None / non-string input -> AttributeError (all three functions)
2. Empty string, missing ':' or '|' -> ValueError on unpack / int() / float()
3. Empty CSV element or non-numeric value -> ValueError
4. Extra separators (a:b:c, a|b|c) -> ValueError on unpack
5. nan / inf / overflow values slip through float() and give nan/inf rent
6. Negative or zero months, negative rent or discount accepted silently
7. duration_average used floor division (3,4 -> 3 instead of 3.5)

Design choice: invalid input returns None and main() prints a friendly message.

Step 2 - Run the buggy application (valid input)

Valid input works, so the bugs only show on unexpected input.

```
$ python self_storage.py
==== Running parse_rental ====
Enter unit:months (e.g. medium:6): Unit and Months: ('Medium', 6.0)

==== Running duration_average ====
Enter rental months as CSV (e.g. 3,6,12): Average Duration: 7

==== Running payable_rent ====
Enter rent|discount (e.g. 2400|200): Payable Rent: 2632.0
```

Step 2 (cont.) - Buggy application on malformed input

A missing ':' crashes the program with an unhandled ValueError.

```
$ python self_storage.py   # malformed input
==== Running parse_rental ====
Enter unit:months (e.g. medium:6): Traceback (most recent call last):
  File "./self_storage.py", line 31, in <module>
    main()
    ~~~~^^
  File "./self_storage.py", line 17, in main
    result = parse_rental(user_input)
  File "./self_storage.py", line 2, in parse_rental
    unit, months = text.split(':')
    ^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
```

Step 4 - pytest + Hypothesis against the buggy script (tests FAIL)

9 of 13 tests fail. Hypothesis found: '', ':', None, and floor-division in the average.

```
$ python -m pytest test_self_storage.py
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-9.1.1, pluggy-1.6.0
rootdir: .
plugins: platformdirs-4.13.5, anyio-4.15.1, hypothesis-6.170.15
collected 13 items
test_self_storage.py FF.FFF.FFF.F.                                       [100%]
=========================== short test summary info ============================
FAILED test_self_storage.py::test_parse_rental_no_crash - ExceptionGroup: Hyp...
FAILED test_self_storage.py::test_parse_rental_any_type_no_crash - ExceptionG...
FAILED test_self_storage.py::test_parse_rental_result_is_clean - ExceptionGro...
FAILED test_self_storage.py::test_duration_average_no_crash - ExceptionGroup:...
FAILED test_self_storage.py::test_duration_average_any_type_no_crash - Except...
FAILED test_self_storage.py::test_duration_average_is_true_mean - assert False
FAILED test_self_storage.py::test_payable_rent_no_crash - ExceptionGroup: Hyp...
FAILED test_self_storage.py::test_payable_rent_any_type_no_crash - ExceptionG...
FAILED test_self_storage.py::test_payable_rent_result_is_clean - ExceptionGro...
======================== 9 failed, 4 passed in 0.78s ===========================

# Falsifying examples Hypothesis found (from full tracebacks):
_____________________________ test_parse_rental_no_crash ______________________________
AssertionError: parse_rental crashed with: ValueError("could not convert string to float: ''")
Failing test case: test_parse_rental_no_crash(
s=':',
AssertionError: parse_rental crashed with: ValueError('not enough values to unpack (expected 2, got 1)')
Failing test case: test_parse_rental_no_crash(
s='',
AssertionError: parse_rental crashed with: AttributeError("'NoneType' object has no attribute 'split'")
Failing test case: test_parse_rental_no_crash(
s=None,
_____________________ test_parse_rental_any_type_no_crash ____________________
AssertionError: parse_rental crashed with: ValueError('not enough values to unpack (expected 2, got 1)')
Failing test case: test_parse_rental_any_type_no_crash(
s='',
AssertionError: parse_rental crashed with: AttributeError("'NoneType' object has no attribute 'split'")
Failing test case: test_parse_rental_any_type_no_crash(
s=None,
_____________________ test_parse_rental_result_is_clean ____________________
Failing test case: test_parse_rental_result_is_clean(
s=':',
Failing test case: test_parse_rental_result_is_clean(
s='',
Failing test case: test_parse_rental_result_is_clean(
s=None,
_____________________ test_duration_average_no_crash ____________________
AssertionError: duration_average crashed with: ValueError("invalid literal for int() with base 10: ''")
Failing test case: test_duration_average_no_crash(
s='',
AssertionError: duration_average crashed with: AttributeError("'NoneType' object has no attribute 'split'")
Failing test case: test_duration_average_no_crash(
s=None,
_________________ test_duration_average_any_type_no_crash _________________
AssertionError: duration_average crashed with: ValueError("invalid literal for int() with base 10: ''")
Failing test case: test_duration_average_any_type_no_crash(
s='',
AssertionError: duration_average crashed with: AttributeError("'NoneType' object has no attribute 'split'")
Failing test case: test_duration_average_any_type_no_crash(
s=None,
_____________________ test_duration_average_is_true_mean ____________________
E   assert False
E   Failing test case: test_duration_average_is_true_mean(
_____________________________ test_payable_rent_no_crash ______________________________
AssertionError: payable_rent crashed with: ValueError("could not convert string to float: ''")
Failing test case: test_payable_rent_no_crash(
s='|',
AssertionError: payable_rent crashed with: ValueError('not enough values to unpack (expected 2, got 1)')
Failing test case: test_payable_rent_no_crash(
s='',
AssertionError: payable_rent crashed with: AttributeError("'NoneType' object has no attribute 'split'")
Failing test case: test_payable_rent_no_crash(
s=None,
_____________________ test_payable_rent_any_type_no_crash ____________________
AssertionError: payable_rent crashed with: ValueError("could not convert string to float: ''")
Failing test case: test_payable_rent_any_type_no_crash(
s='|',
AssertionError: payable_rent crashed with: ValueError('not enough values to unpack (expected 2, got 1)')
Failing test case: test_payable_rent_any_type_no_crash(
s='',
```

Step 4 - pytest + Hypothesis against the buggy script (tests FAIL) (continued)

```
AssertionError: payable_rent crashed with: AttributeError("'NoneType' object has no attribute 'split'")
Failing test case: test_payable_rent_any_type_no_crash(
s=None,
_____________________ test_payable_rent_result_is_clean _____________________
Failing test case: test_payable_rent_result_is_clean(
s='|',
Failing test case: test_payable_rent_result_is_clean(
s='',
Failing test case: test_payable_rent_result_is_clean(
s=None,
FAILED test_self_storage.py::test_duration_average_is_true_mean - assert False
```

Step 5 - Run the fixed application (valid input)

```
$ python self_storage.py
==== Running parse_rental ====
Enter unit:months (e.g. medium:6): Unit and Months: ('Medium', 6.0)

==== Running duration_average ====
Enter rental months as CSV (e.g. 3,6,12): Average Duration: 7.0

==== Running payable_rent ====
Enter rent|discount (e.g. 2400|200): Payable Rent: 2632.0
```

Step 5 (cont.) - Fixed application on malformed input

Bad input now gives a clear message instead of a traceback.

```
$ python self_storage.py   # malformed input
==== Running parse_rental ====
Enter unit:months (e.g. medium:6): Unit and Months: Invalid input (expected unit:months with months > 0)

==== Running duration_average ====
Enter rental months as CSV (e.g. 3,6,12): Average Duration: Invalid input (expected positive whole numbers, comma-
    separated)

==== Running payable_rent ====
Enter rent|discount (e.g. 2400|200): Payable Rent: Invalid input (expected non-negative rent|discount)
```

Step 6 - pytest + Hypothesis against the fixed script (tests PASS)

All 13 tests pass.

```
$ python -m pytest test_self_storage.py -v
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-9.1.1, pluggy-1.6.0 -- /usr/local/bin/python
hypothesis profile 'default'
rootdir: .
plugins: platformdirs-4.13.5, anyio-4.15.1, hypothesis-6.170.15
collecting ... collected 13 items

test_self_storage.py::test_parse_rental_no_crash PASSED                  [  7%]
test_self_storage.py::test_parse_rental_any_type_no_crash PASSED         [ 15%]
test_self_storage.py::test_parse_rental_valid_roundtrip PASSED           [ 23%]
test_self_storage.py::test_parse_rental_result_is_clean PASSED           [ 30%]
test_self_storage.py::test_duration_average_no_crash PASSED              [ 38%]
test_self_storage.py::test_duration_average_any_type_no_crash PASSED     [ 46%]
test_self_storage.py::test_duration_average_between_min_and_max PASSED   [ 53%]
test_self_storage.py::test_duration_average_is_true_mean PASSED          [ 61%]
test_self_storage.py::test_payable_rent_no_crash PASSED                  [ 69%]
test_self_storage.py::test_payable_rent_any_type_no_crash PASSED         [ 76%]
test_self_storage.py::test_payable_rent_never_negative_and_finite PASSED [ 84%]
test_self_storage.py::test_payable_rent_result_is_clean PASSED           [ 92%]
test_self_storage.py::test_known_values PASSED                           [100%]

============================== 13 passed in 1.73s ==============================
```